# 00_LAB_SELECTION_GUIDE

**Authoritative Lab Selection, Completion, and Interview-Positioning Guide**

**By Tobias Weissmann**

Purpose: identify the single required lab, disclose implementation status accurately, and prevent partially implemented reference assets from being mistaken for complete production systems.

## Lab Selection — Read This First

| Lab | Status | Purpose | Buyer should complete? |
|---|---|---|---|
| **Canonical GenAI FDE Interview Lab** | Fully runnable | Core interview and portfolio preparation | **Required** |
| **Advanced Production Architecture Reference** | Partially implemented reference | Study enterprise architecture, safety controls, interfaces, and extension points | **Optional** |
| **Production Lab v1.0** | Deprecated | Historical version retained for reference only | **Do not use** |

Start with and complete the Canonical GenAI FDE Interview Lab. It is the only required hands-on lab in this product. Use the Advanced Production Architecture Reference only after completing the canonical lab or when studying enterprise architecture and implementation extension points. Do not use Production Lab v1.0 for current preparation.

## Classification Standard

| Classification | Meaning |
|---|---|
| **Required / Canonical** | The only lab every learner is expected to run and complete. |
| **Optional / Reference** | A learning resource for broader architecture, controls, interfaces, and extension design. It is not required for completion. |
| **Archived / Deprecated** | Retained only for historical reference. It must not be used as the current study path. |

# 1. Canonical GenAI FDE Interview Lab

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| **Classification** | Required / Canonical |
| **Intended audience** | Learners preparing for GenAI Forward Deployed Engineer, Applied AI Engineer, AI Solutions Engineer, AI Platform Engineer, and closely related interviews. |
| **Learning purpose** | Build evidence of implementation, debugging, evaluation, architecture explanation, tenant-aware request handling, and portfolio storytelling. |
| **Implementation status** | Fully runnable within the documented local environment, subject to the published prerequisites and setup instructions. |
| **Must the learner run it?** | Yes. This is the only required hands-on lab. |
| **Portfolio use** | Yes. It may be used as a portfolio project when the learner can demonstrate their own work, explain limitations, and avoid overstating capabilities. |
| **Known limitations** | Local educational scope; simplified components; not automatically equivalent to a hardened cloud deployment; production integrations may require additional security, persistence, observability, scaling, and operational work. |
| **Recommended sequence** | Use after the Start Here guide and core architecture material. Complete it before opening the advanced reference. |

## 2. Advanced Production Architecture Reference

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| **Classification** | Optional / Reference |
| **Intended audience** | Learners who have completed the canonical lab or who need deeper study of enterprise boundaries, safety controls, provider abstractions, infrastructure patterns, and extension design. |
| **Learning purpose** | Study production-oriented architecture, component contracts, authorization boundaries, deployment patterns, and areas requiring further implementation. |
| **Implementation status** | Partially implemented educational reference. It includes runnable local components, demonstration implementations, extension points, placeholders, and reference infrastructure. |
| **Must the learner run it?** | No. Run only the locally implemented portions when useful; the repository is not a required completion target. |
| **Portfolio use** | Only with precise disclosure. Learners may describe what they studied, extended, implemented, tested, and documented. They must not present unimplemented extension points as completed capabilities. |
| **Known limitations** | External providers, pgvector, migrations, MCP infrastructure, Kubernetes, and Terraform are not complete production services unless separately implemented and verified. |
| **Recommended sequence** | Open only after completing the canonical lab, or when a specific interview topic requires broader production architecture analysis. |

## 3. Production Lab v1.0

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|---|---|
| **Classification** | Archived / Deprecated |
| **Intended audience** | Maintainers or learners who need historical context only. |
| **Learning purpose** | Preserve an earlier implementation for traceability and comparison. |
| **Implementation status** | Deprecated. It is not the current required lab and may contain outdated instructions, architecture, terminology, or incomplete behavior. |
| **Must the learner run it?** | No. |
| **Portfolio use** | No. Use the canonical lab instead. |
| **Known limitations** | Historical content may not match the current product, setup process, quality bar, or interview positioning. |
| **Recommended sequence** | Do not include it in current preparation. Consult only when explicitly comparing versions. |

# Repository Scope and Implementation Status

This repository is an educational production-architecture reference. It demonstrates component boundaries, safety controls, interfaces, and deployment patterns. Some external-provider, database, infrastructure, and MCP implementations are intentionally provided as extension points rather than complete production services.

| Capability | Status | Notes |
|---|---|---|
| FastAPI application | **Implemented** | Runnable locally |
| Tenant-aware request context | **Implemented** | Demonstration implementation |
| In-memory retrieval | **Implemented** | Suitable for local exercises |
| pgvector integration | **Extension point** | Not a complete production adapter |
| OpenAI provider | **Extension point** | Requires user credentials, dependency configuration, and SDK wiring |
| Database migrations | **Reference placeholders** | Not production-ready |
| MCP authorization | **Demonstration only** | Shows contract validation, authorization boundaries, and tenant validation |
| Kubernetes | **Reference manifests** | Requires environment-specific configuration, security review, and hardening |
| Terraform | **Reference structure** | Not a complete deployable environment |

## Status Legend

- Implemented: Included and runnable within the documented local environment.
- Demonstration implementation: Functional for learning but simplified compared with a production system.
- Extension point: Interface, structure, or placeholder provided; additional implementation is required.
- Reference placeholders: Illustrative files or patterns that are not complete operational components.
- Demonstration only: Intended to explain a contract, control, or architecture pattern rather than provide a complete service.
- Reference manifests or structure: Starting material that requires environment-specific implementation and validation.

# Canonical Lab Completion Criteria

A learner should mark the required lab as complete only after satisfying all of the following evidence-based criteria.

☐ The application starts successfully using the documented local setup process.

☐ All required automated tests pass.

☐ At least one document or data-ingestion flow is executed successfully.

☐ At least one retrieval flow returns a valid result.

☐ Tenant context is propagated and enforced in at least one request flow.

☐ At least one evaluation suite is executed and its results are recorded.

☐ At least one failure scenario is intentionally reproduced, diagnosed, and documented.

☐ The learner completes or annotates the system architecture diagram.

☐ The README contains accurate local setup, test, and troubleshooting instructions.

☐ The learner can explain at least three current production limitations and how they would address them.

| Completion criterion | Required evidence |
|---|---|
| **Application startup** | Terminal output, screenshot, or health-check response |
| **Automated tests** | Test command and passing test summary |
| **Ingestion flow** | Sample input and successful ingestion result |
| **Retrieval flow** | Example query and retrieved output |
| **Tenant enforcement** | Test, log, or request showing tenant isolation |
| **Evaluation suite** | Evaluation command, dataset, metrics, and result summary |
| **Failure scenario** | Failure symptoms, diagnosis, root cause, and remediation |
| **Architecture understanding** | Completed or annotated architecture diagram |
| **Documentation quality** | Verified setup and troubleshooting instructions |
| **Production limitations** | Written explanation of three limitations and proposed improvements |

Merely reading the files or starting the application does not constitute completion.

## Learner Decision Flow

1. Complete the Canonical GenAI FDE Interview Lab first.
2. Use it to practise implementation, debugging, evaluation, architecture explanation, and portfolio storytelling.
3. Open the Advanced Production Architecture Reference only when studying broader production boundaries or designing extensions.
4. Do not invest preparation time in Production Lab v1.0.
5. Do not present optional extension points as completed production capabilities during an interview.

# How to Describe the Labs in an Interview

## Canonical lab — recommended wording

I completed a locally runnable GenAI FDE lab covering ingestion, retrieval, tenant-aware request handling, evaluation, testing, and failure analysis. I can explain both the implemented capabilities and the changes required for a production deployment.

## Advanced reference — recommended wording

I studied and extended a production-oriented reference architecture that demonstrates enterprise component boundaries, authorization controls, provider interfaces, deployment patterns, and infrastructure extension points. Some integrations are intentionally represented as contracts or reference structures rather than complete production services.

## Interview integrity warning

Do not claim that pgvector, cloud-provider integration, MCP infrastructure, Kubernetes deployment, or Terraform provisioning is complete unless you personally implemented, tested, and documented those capabilities.

# Claim-Control and Wording Standard

Claims must reflect verified implementation status. Do not perform blind global replacements. Review each title, subtitle, description, feature, architecture section, setup instruction, deployment section, portfolio statement, closing summary, badge, callout, and caption in context.

| Avoid unless proven | Preferred accurate alternative |
|---|---|
| Production-ready | Production-oriented reference architecture |
| Enterprise-grade implementation | Interview-ready implementation patterns |
| Complete pgvector integration | Extension-ready pgvector adapter |
| Fully functional MCP server | MCP safety contract demonstration |
| Complete production deployment | Reference deployment structure |
| Deployable enterprise platform | Production architecture learning environment |
| End-to-end production system | Locally runnable educational implementation |

Reserve “fully runnable” for a component or repository that can be executed successfully through its documented setup instructions and supported by reproducible evidence.

## Ownership and Maintenance Rules

- The Canonical GenAI FDE Interview Lab is the only required lab.
- The Advanced Production Architecture Reference must always be labelled optional and partially implemented.
- Production Lab v1.0 must always be labelled archived or deprecated.
- New required exercises must be added to the canonical lab rather than creating another competing primary lab.
- Any future claim of “implemented,” “runnable,” or “deployable” must be supported by setup instructions, tests, and reproducible evidence.
- README claims must be reviewed whenever implementation status changes.

## README Cross-Reference Requirements

- The canonical README must link to this guide near the top and state: “Required / Canonical — complete this lab first.”
- The advanced README must link to this guide near the top and state: “Optional / Reference — partially implemented.”
- The deprecated README must begin with: “Archived / Deprecated — do not use for current preparation.”
- No README may describe a second repository as canonical.
- Folder order must remain unchanged; use headings, labels, callouts, and cross-references to control the learner path.

## Final Buyer-Facing Rule

Use the canonical lab for hands-on completion and evidence. Use the advanced repository for architecture study and carefully disclosed extensions. Ignore the deprecated lab for current preparation. The product’s credibility depends on making this distinction consistently in every guide, README, badge, and portfolio statement.